



DIGITAL LOGIC DESIGN

Spring 2025

CAR SENSOR SYSTEM

Supervisor: Ma'am Nayyab Khalid

Project Team

Name of Students	Sap ID	Program	Valid Email Address
Amina Batool	62461	BS SE 2	62461@students.riphah.edu.pk
Irfa Arshad	63662	BS SE 2	irfaarshad98@gmail.com
Eman Zahid	63906	BS SE 2	emanzahid.036@gmail.com
Alisha Muqadas	62787	BS SE 2	alishamuqdas185@gmail.com

Date of Submission

20/05/2025

TABLE OF CONTENTS

1.	INTRODUCTION	3
2.	PROBLEM STATEMENT	3
3.	OBJECTIVES	3
4.	SCOPE	3
5.	ABSTRACT	3
6.	COMPONENTS USED	4
7.	WORKING • Module #1 • Module #2	5
8.	LOGIC DIAGRAM • Module #1 • Module #2	6
9.	CODE	7
10.	CIRCUIT DESCRIPTION • Module #2	8
11.	REAL PROJECT IMAGE	9
12.	DRIVE LINK FOR VIDEO	9
13.	RESULTS	9
14.	CONCLUSIONS	10

1. INTRODUCTION

With increasing demand for energy efficiency, automation, and smart infrastructure, the application of digital logic principles in real-life scenarios is rapidly gaining ground. This project integrates three distinct automation modules aimed at improving urban living and safety: an Automatic Smart Street Light System, and an Automatic Car Parking Toll Gate System. Each module reflects practical use-cases using fundamental electronic components, sensors, and microcontrollers

2. PROBLEM STATEMENT

Traditional systems for street lighting, vehicle toll management, and parking safety are inefficient, labor-intensive, and energy-consuming. There is a need for cost-effective, sensor-based automated systems that can improve functionality, reduce manual work, and enhance user experience and safety.

3. OBJECTIVE

- To design cost-effective automated systems that solve real-world problems using digital logic principles.
- To implement sensor-based circuits for energy conservation, smart access control, and safety alerts.
- To demonstrate working prototypes for potential scalability and further enhancements.

4. SCOPE

This project is applicable in:

- **Urban Infrastructure:** Smart lighting for roads and pathways.
- **Parking Management:** Automated toll collection and entry control in parking lots.

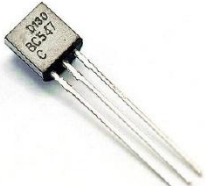
5. ABSTRACT

This project encompasses the development of three independent modules that together illustrate automation in urban infrastructure using digital logic and basic electronic components. The modules include:

- **Automatic Smart Street Light System** – Conserves energy by activating lights only when motion is detected.
- **Automatic Car Parking Toll Gate System** – Automates toll gate operations using an Arduino and sensors.

The implementation involves LDRs, IR sensors, ultrasonic sensors, servo motors, BC547 transistors, buzzers, and Arduino microcontrollers.

6. COMPONENTS USED

Sr No.	Name and Description	
1	LDR (Light Dependent Resistor) for detecting ambient light. It changes the resistance with the variation in the amount of light.	
2	BC-547 Transistor NPN transistor used for switching.	
3	Resistors 10kΩ standard resistors.	 10K ohm
4	IR Proximity Sensor Detects motion or nearby objects.	
5	LEDs Visual indicators for system states. They glow when the circuit conditions are met.	
6	9V Battery Power supply for the circuits.	

7	On/Off Switch Manually controls the power supply to the circuit for ease of testing and demonstration.		
8	Jumper Wires Used to connect all components in the circuit.		
9	Arduino UNO Microcontroller for sensor and motor control.		
10	Ultrasonic Sensor (HC-SR04) Measures distance using sound waves.		
11	Servo Motor (SG90) Operates gate barrier.		
12	Breadboard Solderless circuit prototyping platform.		

7. WORKING

7.1 Module 1: Automatic Smart Street Light

The *Automatic Smart Street Light* system operates based on **motion detection**.

- The LEDs will only turn ON if **motion is also detected**. This is managed by the IR proximity sensor, which checks for movement within a certain range.

- If motion is detected, the circuit allows the LEDs (street lights) to glow, simulating a smart, energy-saving system.

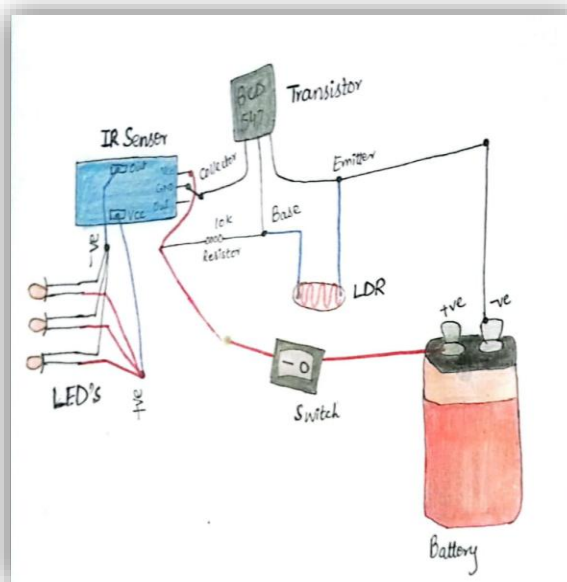
7.2 Module 2: Automatic Car Parking Toll Gate System

The *Automatic Car Parking Toll Gate System* functions using sensor-based automation to detect vehicles and control the gate. The system functions by continuously monitoring the distance between the ultrasonic sensor and any approaching object (vehicle). Here's how it works:

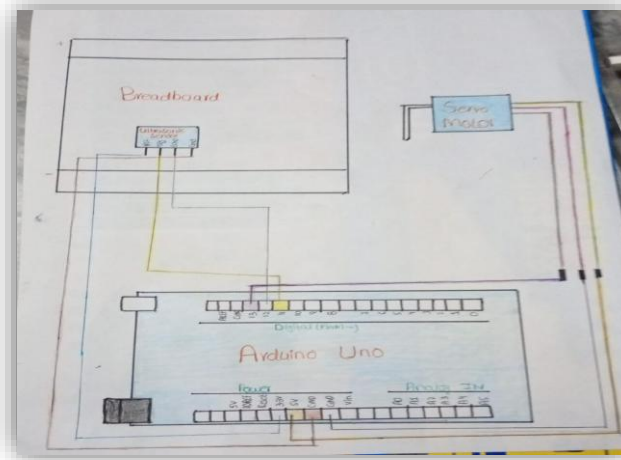
- When powered on, the Arduino initializes the servo motor to position 180° (gate open).
- The ultrasonic sensor, connected to pins 11 (Trig) and 12 (Echo), sends out sound waves and waits for the echo to return.
- The time taken by the echo is used to calculate the distance of the object.
- If a vehicle is detected **within 25 cm**, the servo motor remains at 180° (gate stays open), allowing the vehicle to pass.
- If **no vehicle is detected** (i.e., distance is more than 25 cm), the servo moves to 90° (gate closes).
- The sensor checks every half second (500ms), updating the gate's state in real-time based on vehicle presence.

8. LOGIC DIAGRAMS

8.1 Module #1: Automatic Smart Street Light



8.2 Module #2: Automatic Car Parking Toll Gate System



9. CODE

```
#include <Servo.h>
Servo servo;
int trigPin = 11;
int echoPin = 12;

// Define variables
long duration;
float distance;

void setup() {
  Serial.begin(9600); // Initialize Serial Monitor
  servo.attach(13);
  servo.write(180);
  delay(2000);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // Clears the trigPin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  // Sets the trigPin on HIGH state for 10 microseconds
```

```

digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);

// Reads the echoPin, returns the sound wave travel time in microseconds
duration = pulseIn(echoPin, HIGH, 20000); // Timeout added

// Calculate the distance
distance = duration * 0.034 / 2;

// Print the distance on the Serial Monitor
Serial.print("Distance: ");
Serial.println(distance);

if (distance <= 25) { // Modify based on sensor placement
    servo.write(180);
    delay(3000);
} else {
    servo.write(90);
}
delay(500); // Prevent excessive printing
}

```

10. CIRCUIT DESCRIPTION

10.1 Module #1:

- **Ultrasonic Sensor (HC-SR04):**
 - **Trig pin** is connected to **pin 11** on the Arduino.
 - **Echo pin** is connected to **pin 12**.
 - The Arduino sends a 10µs HIGH pulse to the Trig pin, which causes the sensor to emit an ultrasonic burst.
 - It then listens on the Echo pin for the returning signal. The time is measured using pulseIn() and converted into distance.
- **Servo Motor (SG90):**
 - Connected to **pin 13** on the Arduino.
 - Initially set to **180°** (gate open).
 - If the sensor detects no vehicle, it moves to **90°** (gate closed).

- The motor holds the gate open for 3 seconds (delay(3000)) when a vehicle is detected.
- **Arduino Uno:**
 - Handles all logic and control.
 - Reads input from the sensor and drives the servo motor accordingly.
 - Prints real-time distance values to the **Serial Monitor** for debugging and monitoring.

11.REAL PROJECT IMAGE



12.DRIVE LINK FOR VIDEO

[https://drive.google.com/file/d/1zzrQSzyD-3wIL7z0GFH8Q2-Zy42sbiHm/view?usp=drive link](https://drive.google.com/file/d/1zzrQSzyD-3wIL7z0GFH8Q2-Zy42sbiHm/view?usp=drive_link)

13.RESULTS

All three modules performed their intended functions successfully during testing.

- Module 1 conserved energy by activating lights only on motion detection.
- Module 2 automated toll gate operation, reducing manual labor and improving vehicle flow.

14.CONCLUSIONS

This project successfully demonstrates how simple digital and analog components can be used to build functional, real-world automation systems. The smart street lighting system promotes energy conservation; the automated parking gate improves traffic management. These modules serve as practical examples of how embedded systems can improve urban efficiency and quality of life, while remaining affordable and accessible for student and entry-level engineers. Future upgrades can include integration with wireless technologies, cloud-based data logging, and AI-driven decision-making.